Chicago Crimes Analysis Pipeline

End-to-End Big Data Analytics for Public Safety

Yazan Kbaili Ali Salloum

Innopolis University – Big Data Course

May 2026

Agenda

1. Business Problem & Objectives
2. Dataset Overview
3. Pipeline Architecture
4. Stage I – Data Collection & Ingestion
5. Stage II – Hive Optimizations & EDA
6. Stage III – Predictive ML Pipeline
7. Risk Assessment
8. Challenges Faced
9. Future Prospects
10. Team Participation
11. Conclusion
12. Dashboard Demonstration

Business Problem

Problem:

- Chicago PD needs data-driven patrol resource allocation
- 8.5M+ crime records over 25 years – manual analysis infeasible
- Need to identify high-risk spatiotemporal patterns

Business Objective:

- Predict arrest likelihood from incident characteristics
- Enable targeted patrol deployment

Key Metric

Business-AI Link: Higher AUC-PR $\rightarrow$ Better arrest prediction $\rightarrow$ Smarter resource allocation $\rightarrow$ Improved clearance rates

Dataset – Chicago Crimes (2001–2026)

Attribute	**Value**
Source	Chicago Data Portal
Records	8,549,662
Columns	22
File Size	1.9 GB (CSV)
Target Variable	`Arrest` (binary)
Arrest Rate (TRUE)	25.09%
Arrest Rate (FALSE)	74.91%
Time Span	2001 – 2026

Challenge

Class imbalance: only 25% positive class – requires careful metric selection

Pipeline Architecture

Data Flow

Socrata API → PostgreSQL → Sqoop (Parquet+Snappy) → HDFS → Hive → PySpark ML → Superset

Node	Services
hadoop-01	Spark, Hive, Sqoop, JupyterHub
hadoop-02	HDFS NameNode, Hive Metastore
hadoop-03	HiveServer2, YARN RM, Superset
hadoop-04	PostgreSQL 14, HDFS DataNode

Stage I: Data Collection & Ingestion

Process:

1. Download from Socrata API → 1.9 GB CSV
2. Load into PostgreSQL via `copy_expert` – 8,549,662 rows
3. Sqoop import to HDFS – 3 format/codec benchmarks
4. Hive external table creation

Sqoop Benchmark Results:

Format	Size	Time
Parquet + Snappy	509 MB	54s ★
AVRO + Snappy	845 MB	84s
Parquet + Gzip	360 MB	93s

Winner: Parquet + Snappy – best balance of compression and speed

Stage II: Hive Optimizations

Partitioning:

- Partition key: `year`
- 26 partitions (2001–2026)
- Enables time-range pruning

Bucketing:

- Bucket key: `district`
- 22 buckets
- Enables efficient joins

Optimized Table

Property	**Value**
Rows	8,549,662
Partitions	26
Buckets	22
Files	370

Stage II: EDA – 7 Key Insights

#	Insight	Key Finding
Q1	Arrest rate by district	Dist. 11: 40.89% vs Dist. 16: 18.19%
Q2	Top 15 crime types	THEFT (1.82M), BATTERY (1.56M)
Q3	Crimes by hour	Peaks at 12:00 and 18:00
Q4	Arrest rate trend	29.21% (2001) $\rightarrow$ 12–16% (2020s)
Q5	Crimes by month	Seasonal summer peaks
Q6	Arrest by community area	Wide variation across 77 areas
Q7	Domestic vs non-domestic	Non-domestic (26.3%) $>$ domestic (19.2%)

Stage III: Feature Engineering

Custom PySpark Transformers:

- **SinCosTransformer:** Cyclical encoding of time features
  - hour_of_day → hour_sin, hour_cos (period=24)
  - day_of_week → day_sin, day_cos (period=7)
  - month → month_sin, month_cos (period=12)
- **GeoToECEFTransformer:** Geospatial encoding
  - lat/lon → ECEF x, y, z (WGS84 ellipsoid)

Pipeline Stages:

1. SinCos + ECEF transformers
2. StringIndexer + OneHotEncoder (4 categoricals)
3. VectorAssembler → VarianceThresholdSelector
4. Train/Test split: 80/20 (801k / 200k records)

Stage III: Models & Hyperparameter Tuning

Random Forest

- Grid: `numTrees` [20, 50]
- Grid: `maxDepth` [5, 10]
- 3-fold CV
- Metric: AUC-PR
- Best: numTrees=50, maxDepth=10

Gradient Boosted Trees

- Grid: `maxDepth` [3, 6]
- Grid: `stepSize` [0.05, 0.1]
- 3-fold CV
- Metric: AUC-PR
- Best: maxDepth=6, stepSize=0.1

Why AUC-PR?

Dataset is imbalanced (25% arrest rate). AUC-PR focuses on positive class performance.

Stage III: Results & Model Comparison

Model	**AUC-ROC**	**AUC-PR**	**Accuracy**	**F1**
Logistic Regression (pilot)	0.6529	0.3813	0.7491	0.6666
Random Forest (tuned)	0.8738	0.7968	0.8492	0.8256
GBT (tuned) ★	**0.8844**	**0.8176**	**0.8809**	**0.8710**

Winner: Gradient Boosted Trees

- Highest AUC-PR (0.8176) – critical for imbalanced data
- Highest F1 (0.8710) – best precision/recall balance
- Optimal hyperparams: maxDepth=6, stepSize=0.1

Stage II Pilot vs Stage III Final

Metric	Stage II (10%)	Stage III (1M)	Change
RF AUC-PR	0.8352	0.7968	-0.0384
GBT AUC-PR	0.8418	0.8176	-0.0242
RF F1	0.8742	0.8256	-0.0486
GBT F1	0.8817	0.8710	-0.0107

- Slight decrease expected: Stage III uses larger, more diverse sample
- **Key insight:** GBT degrades less than RF – more robust to data diversity

Risk Assessment & Model Limitations

Potential Risks:

1. **Temporal drift:** Crime patterns evolve; model trained on 2001–2026 may not generalize to future years
2. **Geographic bias:** Some districts have very few records – predictions may be unreliable
3. **Class imbalance:** Model may over-predict "no arrest" (74.9% majority class)

Mitigation Strategies:

- Periodic retraining with new data
- District-specific threshold tuning
- AUC-PR optimization (already addresses class imbalance)
- Monitor prediction distribution for drift detection

Challenges Faced

- **Cluster resource limits:** YARN queue contention with other teams
- **Date format conversion:** Sqoop stores timestamps as BIGINT epoch – required `from_unixtime()` conversion
- **Hive strict mode:** Dynamic partitioning required `nonstrict` mode configuration

Future Prospects

- **Deep Learning:** Explore MultilayerPerceptronClassifier for non-linear patterns
- **Real-time pipeline:** Stream crime data via Kafka + Spark Streaming
- **Model interpretation:** SHAP/LIME for explaining individual predictions
- **Geospatial clustering:** DBSCAN on crime hotspots for patrol zone optimization
- **Feature expansion:** Weather data, census demographics, school calendar integration
- **Active learning:** Reduce labeling cost for new crime types

Team Participation

Member	Contribution	%
Yazan Kbaili	Hive optimizations (partitioning/bucketing) 7 EDA insights via PySpark ML pipeline (RF + GBT with tuning) Superset dashboard (Stage IV) Presentations and reports	50%
Ali Salloum	Data collection (Socrata API) PostgreSQL schema and loading Sqoop import and benchmarks Hive external tables Pipeline automation scripts	50%

Conclusion

Key Achievements

- Built end-to-end big data pipeline on 4-node Hadoop cluster
- Processed 8.5M crime records across 25 years
- Discovered 7 actionable EDA insights
- Trained and compared 2 ML models with hyperparameter tuning
- Published interactive Superset dashboard

Best Model

Gradient Boosted Trees – AUC-PR: 0.8176, F1: 0.8710

Thank you! Questions?

Superset Dashboard Demonstration

My colleague Ali will now present the live dashboard.

Three tabs:

- Data Description
- EDA Insights (7 charts)
- ML Modeling